

ERRATUM:

Controllability of Nonlinear Discrete-Time Systems: A Lie-Algebraic Approach*

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As stated, Corollary 5.3 is false. The indices are missing in the statement. The correct version is as follows.

COROLLARY 5.3. *If an analytic system with connected $\mathcal{U}$ is forward accessible from x in k steps (that is, $\text{int } A_k^+(x) \neq \emptyset$), then*

$$\dim L_k^-(y) = n \quad \text{for any } y \in A_k^+(x).$$

Similarly, if it is backward accessible from x in k steps, then

$$\dim L_k^+(y) = n \quad \text{for any } y \in A_k^-(x).$$

Proof. The first statement follows directly from the first inclusion in Proposition 5.2 and the inclusion

$$\bigcup_{k>1} \text{Orb}_{\Delta_k^+}(x) \subset \text{Orb}_L(x).$$

The second statement follows analogously. $\square$

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